

Software Verification and Validation

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Binary Relations

Definition

A binary relation R between A and B is an element of $\text{Pow}(A \times B)$, i.e.,

$$R \subseteq A \times B$$

Examples

$$\text{Parent} \subseteq \text{Person} \times \text{Person}$$

$$\text{Parent} = \{(John, Autumn), (John, Sam)\}$$

$$\text{Square} \subseteq \mathbb{Z} \times \mathbb{N}$$

$$\text{Square} = \{(1, 1), (-1, 1), (-2, 4)\}$$

Domain and Image of Relations

Definition Domain

- The set of first elements in the relation

$$\text{domain}(\text{Parent}) = \{John\}$$

Image

- The set of second elements in the relation

$$\text{image}(\text{Square}) = \{1, 4\}$$

Important

The domain and image contain only the elements that actually appear in the relation.

Exercise: Domain and Image

Given the relation:

$$R = \{(1, \text{blue}), (2, \text{blue}), (1, \text{red})\}$$

Find:

- $\text{Domain}(R)$
- $\text{Image}(R)$

Answer

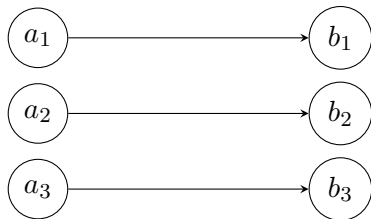
$$\text{domain}(R) = \{1, 2\}$$

$$\text{image}(R) = \{\text{blue}, \text{red}\}$$

One-to-One Relation

Definition

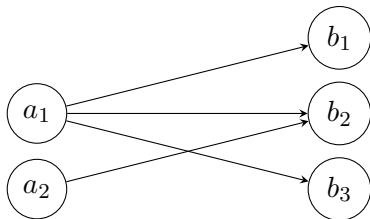
Each element in set A is related to exactly one unique element in set B .



One-to-Many Relation

Definition

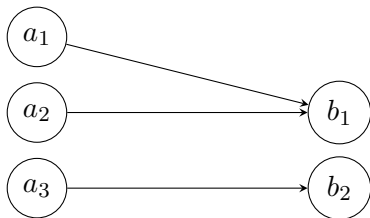
One element in set A can be related to multiple elements in set B .



Many-to-One Relation

Definition

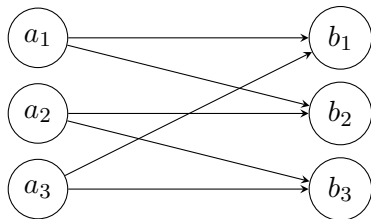
Multiple elements in set A can be related to the same element in set B .



Many-to-Many Relation

Definition

Multiple elements in set A can be related to multiple elements in set B .



Functions as Relations

Definition

- A function is a relation of arity $n + 1$
- No two distinct tuples may have the same first n elements

For Binary Relations ($n = 1$)

$$\forall (a_1, b_1) \in F, \forall (a_2, b_2) \in F,$$

$$a_1 = a_2 \Rightarrow b_1 = b_2$$

Key Idea

Each input maps to exactly one output.

Which of These Are Functions?

1. Parent Relation

$$\text{Parent} = \{(John, Autumn), (John, Sam)\}$$

Not a function

- John maps to two different values

2. Square Relation

$$\text{Square} = \{(1, 1), (-1, 1), (-2, 4)\}$$

Function

- Each input has exactly one output

Which of These Are Functions? (Cont.)

3. ClassGrades Relation

$$\text{ClassGrades} = \{(Todd, A), (Jane, B)\}$$

Function

- Each person has exactly one grade

Important

A function is a relation that is **many-to-one** or **one-to-one**, but never one-to-many.

Special Kinds of Functions

Consider a function

$$f : S \rightarrow T$$

where:

- S is the domain
- T is the codomain

Total Function

- f is **total** if it is defined for every element of S

$$\forall x \in S, f(x) \text{ is defined}$$

Partial Function

- f is **partial** if it is undefined for some elements of S

$$\exists x \in S, f(x) \text{ is undefined}$$

Special Kinds of Functions

Consider a function

$$f : S \rightarrow T$$

Injective (One-to-One)

- f is injective if no two different elements in S map to the same element in T

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Surjective (Onto)

- f is surjective if every element of T is mapped by some element in S

$$\forall y \in T, \exists x \in S \text{ such that } f(x) = y$$

Bijjective Function

- A function is bijective if it is both:
 - Injective (one-to-one)
 - Surjective (onto)

Key Property

- Every element in S maps to a unique element in T
- Every element in T is mapped by exactly one element in S

Key Insight

Bijjective functions establish a perfect pairing between two sets.

Relation Composition

Idea

- Combine two relations to produce a new relation
- Map the domain of the first relation to the image of the second

Definition

Given:

$$s \subseteq A \times B, \quad r \subseteq B \times C$$

their composition is:

$$s; r \subseteq A \times C$$

Meaning

$$(a, c) \in s; r$$

iff there exists some b such that:

$$(a, b) \in s \wedge (b, c) \in r$$

Example of Relation Composition

Given:

$$s = \{(red, 1), (blue, 2)\}, \quad r = \{(1, 2), (2, 4), (3, 6)\}$$

Composition

$$s; r = \{(red, 2), (blue, 4)\}$$

Explanation

- $red \rightarrow 1$ in s
- $1 \rightarrow 2$ in r
- Therefore:

$$red \rightarrow 2$$

Key Insight

Relation composition chains relations together through shared intermediate values.

Relation Closure

Intuition

- The closure of a relation repeatedly follows the relation
- Continue navigating through the relation until no new connections can be found

Definition

For a relation:

$$r \subseteq S \times S$$

its transitive closure, written r^+ , is:

$$r^+ \equiv r \cup (r; r) \cup (r; r; r) \cup \dots$$

Meaning

- Direct connections
- Indirect connections through intermediate elements

Examples of Relation Closure

Grandparent Relation

$$\text{GrandParent} = \text{Parent}; \text{Parent}$$

- Follow the Parent relation twice

Ancestor Relation

$$\text{Ancestor} = \text{Parent}^+$$

- Parent
- Grandparent
- Great-grandparent
- And so on...

Relation Transpose

Intuition

- The transpose of a relation reverses all ordered pairs

For a relation:

$$r \subseteq S \times T$$

its transpose, written $\sim r$, is:

$$\sim r \equiv \{(b, a) \mid (a, b) \in r\}$$

Meaning

- Swap the direction of the relation

Examples of Relation Transpose

Child Relation

$$\text{ChildOf} = \sim \text{Parent}$$

- Reverse the Parent relation

Descendant Relation

$$\text{DescendantOf} = (\sim \text{Parent})^+$$

- Child
- Grandchild
- Great-grandchild
- And so on...

Thank You

Thank you for your attention!

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